MA3521 — A Forced 3×3 IVP

Name:

Solve the IVP completely and determine its leading behavior as $t \to \infty$

$$\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-4\\-4\\-4\end{pmatrix} + t\begin{pmatrix}2\\2\\0\end{pmatrix} \qquad A = \begin{pmatrix}-4 & 2 & 4\\-2 & 1 & 3\\-4 & 3 & 3\end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix}3\\1\\4\end{pmatrix}$$

1a. Eigenvalues

$\det(A - \lambda I) =$ $= 0$

$\lambda_+ =$ $\lambda_- =$ $\lambda_3 =$

Use $\mathrm{Im}(\lambda_+) > 0$ **1b. Eigenvectors**

Row reduce $[A - \lambda_+ I \mid \mathbf{0}]$ and find $\mathbf{v}_+$ with third entry 1

$\mathbf{v}_+ =$ $\mathbf{v}_- = \overline{\mathbf{v}_+} =$

Row reduce $[A - \lambda_3 I \mid \mathbf{0}]$ and find $\mathbf{v}_3$ with first entry 1

$\mathbf{v}_3 =$

2a. Real complementary solution

Write $\mathbf{v}_+ = \mathbf{p} + i\mathbf{q}$ with real vectors $\mathbf{p}, \mathbf{q}$

$\mathbf{p} =$ $\mathbf{q} =$

Expand $e^{\lambda_+ t}\mathbf{v}_+$ and separate its real and imaginary parts

$\mathbf{y}_1(t) = \mathrm{Re}(e^{\lambda_+ t}\mathbf{v}_+) =$

$\mathbf{y}_2(t) = \mathrm{Im}(e^{\lambda_+ t}\mathbf{v}_+) =$

$\mathbf{y}_3(t) = e^{\lambda_3 t}\mathbf{v}_3 =$

$\mathbf{y}_c(t) = \sum_{k=1}^{3} c_k \mathbf{y}_k(t) =$

2b. Exponential particular solution

For $\mathbf{f}_1(t) = e^{-2t}(-4, -4, -4)^T$ try $\mathbf{y}_{p_1}(t) = \mathbf{a}e^{-2t}$

$\mathbf{y}'_{p_1}(t) =$

Substitute into $\mathbf{y}'_{p_1} = A\mathbf{y}_{p_1} + \mathbf{f}_1$ and cancel the exponential

$\mathbf{a} =$ $\mathbf{a} =$

$\mathbf{y}_{p_1}(t) =$

3a. Linear particular solution

For $\mathbf{f}_2(t) = t(2,2,0)^T$ try $\mathbf{y}_{p_2}(t) = \mathbf{b}t + \mathbf{d}$

$\mathbf{y}'_{p_2}(t) =$

Substitute into $\mathbf{y}'_{p_2} = A\mathbf{y}_{p_2} + \mathbf{f}_2$ and collect powers of t

Match the coefficient of t and the constant term

$A\mathbf{b} =$ $A\mathbf{d} =$

Solve these linear systems

$\mathbf{b} =$ $\mathbf{d} =$

$\mathbf{y}_{p_2}(t) =$

3b. Particular and general solutions

$\mathbf{y}_p(t) = \mathbf{y}_{p_1}(t) + \mathbf{y}_{p_2}(t) =$

$\mathbf{y}(t) = \mathbf{y}_c(t) + \mathbf{y}_p(t) =$

4a. Initial condition

$\mathbf{y}_p(0) =$ $\mathbf{y}_0 - \mathbf{y}_p(0) =$

Write $M\mathbf{c} = \mathbf{y}_0 - \mathbf{y}_p(0)$ with $M = [\mathbf{y}_1(0)\ \mathbf{y}_2(0)\ \mathbf{y}_3(0)]$

$M =$ $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

Form the augmented matrix and row reduce

$\mathbf{c} =$

4b. IVP solution and leading behavior

Substitute the constants and both particular contributions

$\mathbf{y}(t) =$

Identify the term that dominates as $t \to \infty$

$\mathbf{y}(t) \sim$ as $t \to \infty$